

Self Assessment Quiz

Question ①

$$\begin{aligned} \textcircled{1} \quad 3 \begin{bmatrix} -4 & 3 \\ -2 & 4 \end{bmatrix} - 5 \begin{bmatrix} 5 & -3 \\ 2 & 4 \end{bmatrix} &= \begin{bmatrix} -12 & 9 \\ -6 & 12 \end{bmatrix} - \begin{bmatrix} 25 & -15 \\ 10 & 20 \end{bmatrix} \\ &= \begin{bmatrix} -12-25 & 9+15 \\ -6-10 & 12+20 \end{bmatrix} \\ &= \begin{bmatrix} -37 & 24 \\ -16 & -8 \end{bmatrix} \end{aligned}$$

$$\begin{bmatrix} -37 & 24 \\ -16 & -8 \end{bmatrix} \pmod{3} = \begin{bmatrix} 2 & 0 \\ 2 & 1 \end{bmatrix}$$

Question 2

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 1 & 2 \\ 1 & 0 & 2 \end{bmatrix}$$

$$2 \cdot 1 + 1 \cdot 1 + 1 \cdot 1 = 4$$

$$1 \cdot 1 + 2 \cdot 1 + 1 \cdot 0 = 3$$

$$2 \cdot 1 + 1 \cdot 1 + 0 = 3$$

$$1 \cdot 1 + 2 \cdot 1 + 0 \cdot 0 = 3$$

$$2 \cdot 2 + 1 \cdot 2 + 1 \cdot 2 = 8$$

$$1 \cdot 2 + 2 \cdot 2 + 0 = 6$$

$$\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 0 \\ 0 & 1 & 0 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 4 & 3 & 8 \\ 3 & 3 & 6 \\ 1 & 1 & 2 \\ 3 & 3 & 6 \end{bmatrix}$$

$$0 + 1 \cdot 1 + 0 = 1$$

$$1 \cdot 2 + 2 \cdot 2 + 0 = 6$$

$$0 + 1 \cdot 1 + 0 = 1$$

$$1 \cdot 1 + 2 \cdot 1 + 0 = 3$$

$$0 + 1 \cdot 2 + 0 = 2$$

$$1 \cdot 1 + 2 \cdot 1 + 0 = 3$$

$$AB = \begin{bmatrix} 4 & 3 & 8 \\ 3 & 3 & 6 \\ 1 & 1 & 2 \\ 3 & 3 & 6 \end{bmatrix}$$

$$AB \pmod{3} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \\ 1 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

!

I've accidentally written this as 1 in NUMBAS.

Question 3

$$\textcircled{3} \quad A = \begin{bmatrix} 4 & -2 & 2 \\ 1 & -1 & -4 \\ -4 & -3 & 4 \end{bmatrix}$$

$$\begin{aligned} \det(A) &= -4 \times C_{31} + -3 \times C_{32} + 4 \times C_{33} \\ &= -4 \times 1 \times \det \begin{pmatrix} -2 & 2 \\ -1 & -4 \end{pmatrix} + -3 \times -1 \times \det \begin{pmatrix} 4 & 2 \\ 1 & -4 \end{pmatrix} + 4 \times 1 \times \det \begin{pmatrix} 4 & -2 \\ 1 & -1 \end{pmatrix} \\ &= -4(8+2) + 3(-16-2) + 4(-4+2) \\ &= -4 \times 10 + 3 \times -18 + 4 \times -2 \\ &= -102 \end{aligned}$$

$$-102 \equiv 6 \pmod{18}$$

$$18 \times 6 = 102 \Rightarrow 6$$

Question ④

$$A = \begin{pmatrix} 8 & 9 & 3 \\ 5 & 5 & -2 \\ -6 & 0 & -7 \end{pmatrix}$$

$$C_{11} = (-1)^{1+1} \times \det \begin{pmatrix} 5 & -2 \\ 0 & -7 \end{pmatrix} = 1 \times (-35 + 0) = -35$$

$$C_{12} = (-1)^{1+2} \times \det \begin{pmatrix} 5 & -2 \\ -6 & -7 \end{pmatrix} = -1 \times (5 \times -7 - 6 \times 2) = 47$$

$$C_{13} = (-1)^{1+3} \times \det \begin{pmatrix} 5 & 5 \\ -6 & 0 \end{pmatrix} = 1 \times (0 + 5 \times 6) = 30$$

$$C_{21} = (-1)^{2+1} \times \det \begin{pmatrix} 9 & 3 \\ 0 & -7 \end{pmatrix} = -1 \times (9 \times -7 - 0) = 63$$

$$C_{22} = (-1)^{2+2} \times \det \begin{pmatrix} 8 & 3 \\ -6 & -7 \end{pmatrix} = 1 \times (8 \times -7 + 6 \times 3) = -38$$

$$C_{23} = (-1)^{2+3} \times \det \begin{pmatrix} 8 & 9 \\ -6 & 0 \end{pmatrix} = -1 \times (8 \times 0 + 9 \times 6) = -54$$

$$C_{31} = (-1)^{3+1} \times \det \begin{pmatrix} 9 & 3 \\ 5 & -2 \end{pmatrix} = 1 \times (9 \times -2 - 15) = -33$$

$$C_{32} = (-1)^{3+2} \times \det \begin{pmatrix} 8 & 3 \\ 5 & -2 \end{pmatrix} = -1 \times (-16 - 15) = 31$$

$$C_{33} = (-1)^{3+3} \times \det \begin{pmatrix} 8 & 9 \\ 5 & 5 \end{pmatrix} = +1 \times (8 \times 5 - 9 \times 5) = -5$$

$$\therefore C(A) = \begin{bmatrix} -35 & 47 & 30 \\ 63 & -38 & -54 \\ -33 & 31 & -5 \end{bmatrix}$$

$$\det(A) = 8 \times C_{11} + 9 \times C_{12} + 3 \times C_{13}$$

$$= 8 \times -35 + 9 \times 47 + 3 \times 30$$

$$= 233$$

$$\text{adj } A = \begin{bmatrix} -35 & 63 & -33 \\ 47 & -38 & 31 \\ 30 & -54 & -5 \end{bmatrix}$$

$$A^{-1} = \frac{1}{233} \text{adj } A$$

$$A^{-1} = \frac{1}{233} \begin{bmatrix} -35 & 63 & -33 \\ 47 & -38 & 31 \\ 30 & -54 & -5 \end{bmatrix} = \begin{bmatrix} -35/233 & 63/233 & -33/233 \\ 47/233 & -38/233 & 31/233 \\ 30/233 & -54/233 & -5/233 \end{bmatrix} = \begin{bmatrix} -0.15 & 0.27 & -0.14 \\ 0.2 & -0.16 & 0.13 \\ 0.13 & -0.23 & -0.23 \end{bmatrix}$$

$$233 \equiv 1 \pmod{2} \Rightarrow (\det A)^{-1} \pmod{2} = 1$$

$$A^{-1} = \frac{1}{233} \begin{bmatrix} -35 & 63 & -33 \\ 47 & -38 & 31 \\ 30 & -54 & -5 \end{bmatrix} \equiv 1 \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \pmod{2}$$

SIT281 Matrix Algebra

Click on a question number to see how your answers were marked and, where available, full solutions.

Question Number	Score	
Question 1	10 / 10	Review
Question 2	3 / 6	Review
Question 3	14 / 14	Review
Question 4	14 / 14	Review
Total	41 / 44 (93%)	

from that type

Performance Summary

Exam Name:	SIT281 Matrix Algebra
Session ID:	0796322395
Exam Start:	Sat Jul 25 2026 10:34:16
Exam Stop:	Sat Jul 25 2026 11:40:54
Time Spent:	1:06:37

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